

X Bot Detector: A Real-Time Browser-Based Approach

Data Science Capstone Project Presentation

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How Much of the Internet is Fake?

32%

Bad Bot Traffic

Malignant automated traffic in 2023

49%

Total Automated

Combined bot traffic across all
categories

1/3

Not Human

Accounts you encounter probably aren't
people

Source: Imperva 2024 Bad Bot Report

Why Bot Detection Matters



Amplification

Disproportionately spread
low-credibility news



Political Influence

Distorted 2016 US presidential
discourse



Inflammation

Boost exposure to negative,
inflammatory content



Scale

Makes up significant fraction of social media activity



Deception

Increasingly indistinguishable from humans

Existing Tools Fall Short...

What We Have

- Botometer requires X's API access (paywalled since 2023)
- Must leave platform and visit separate site
- Rate-limited, slow on-demand checks
- Botwiki catalogs known bots, not live detection
- Manual lookup with no real-time scoring

What We Need

- Runs inside browser without leaving X
- No API keys, signup, or payment
- One-click profile scoring
- Readable explanation for every result
- Real-time detection on suspicious accounts

Project Aim & Success Criteria

Build a machine-learning bot detector that scores any X profile and comments from a comment thread in real time, from inside the user's browser.

01

F1 Score > 0.80

On the held-out test data

02

Fast Performance

Live scoring in Chrome, under 1 second per profile

03

Explainable AI

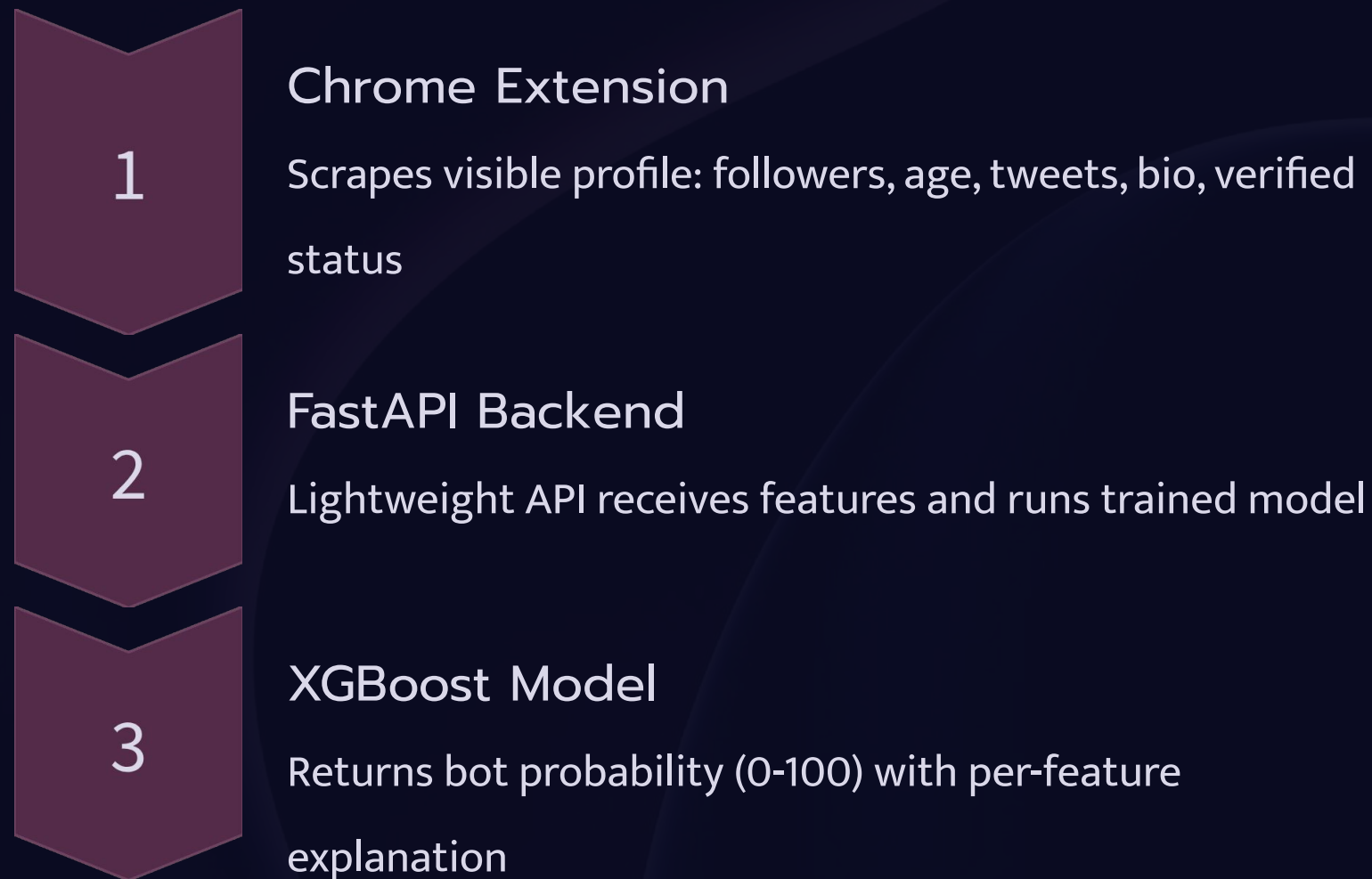
Every prediction comes with readable explanation

04

Real-World Valid

Works on current profiles and threads, not just training data

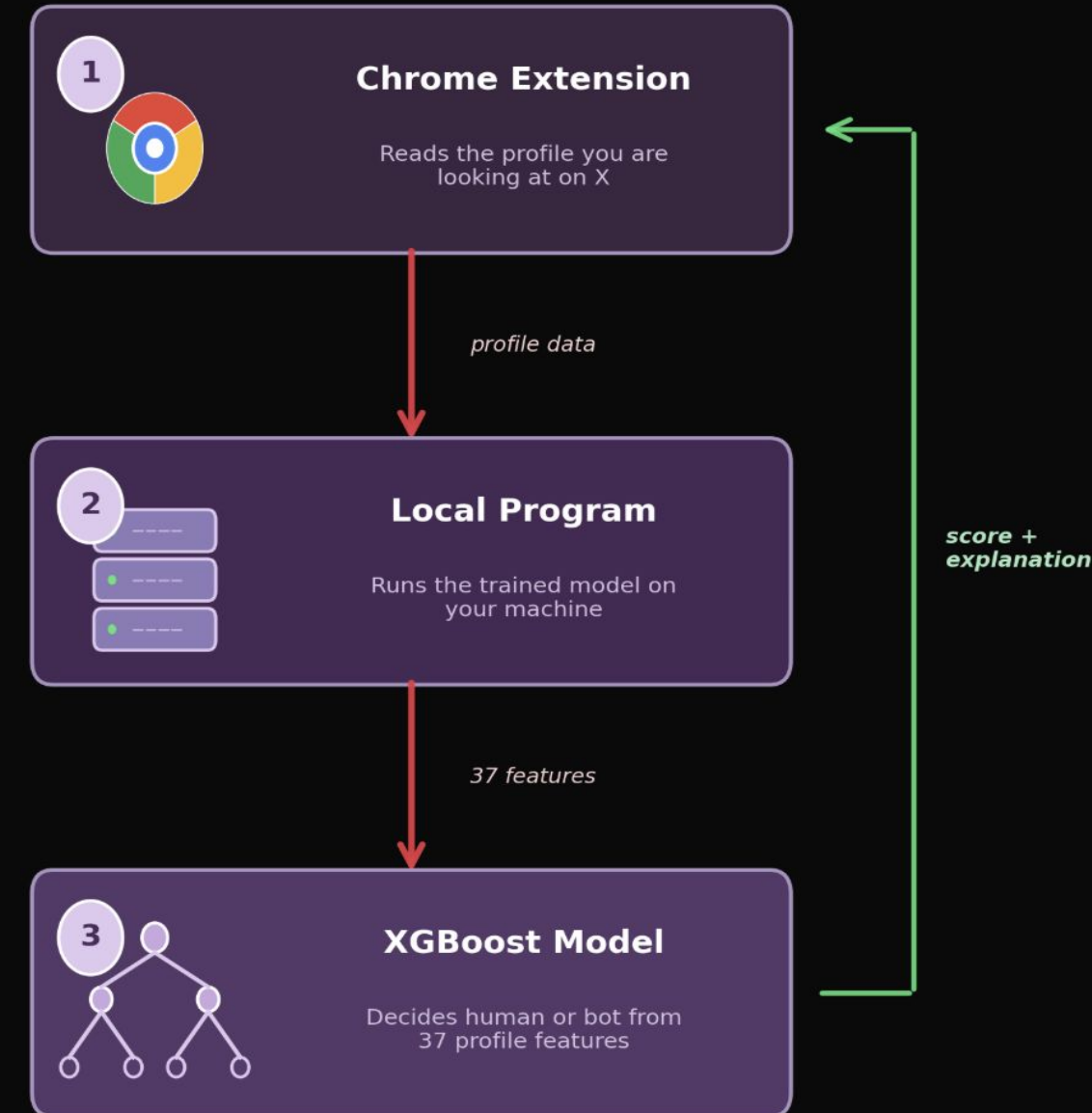
System Architecture



Round-trip time: under 1 second. The model runs the XGBoost classifier on 37 engineered features derived from publicly visible profile metadata.

How the system works

Round-trip time: under 1 second



Everything runs on your own machine.

No cloud. No data leaves the browser.

Dataset Overview

37,438

Total Accounts
Expertly labeled profiles

25,013

Human Accounts
Real users in training set

12,425

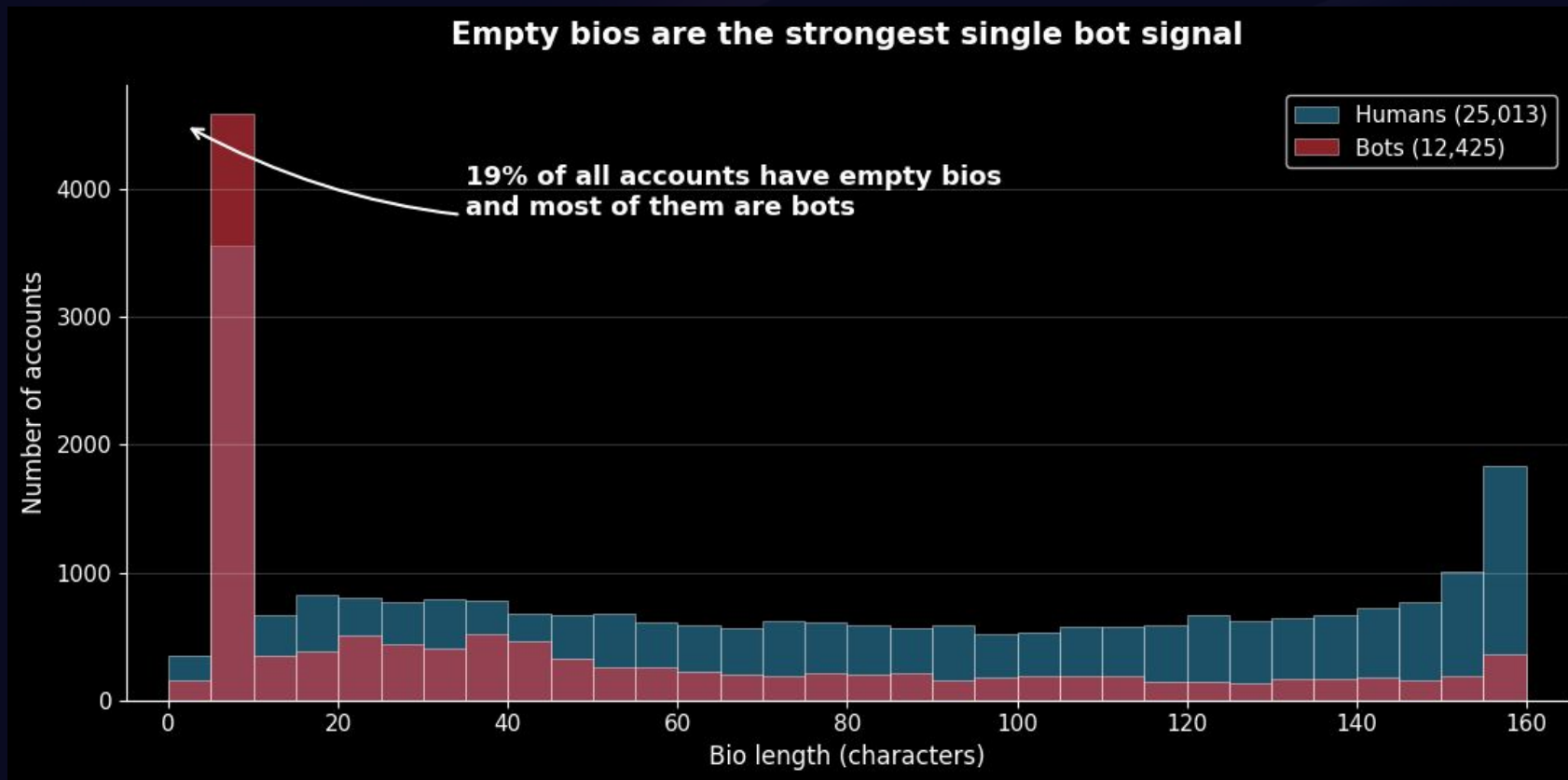
Bot Accounts
Malicious automated profiles

2:1

Class Ratio
Humans to bots split

What Do Bots Look Like?

Key Indicators



19% have empty bios , overwhelmingly bots cluster at zero length

- Newer accounts with recent creation dates
- Inhumanly fast posting (>100 tweets/day in tail)
- Low profile completeness: no picture, location, or header image
- Screen names with random digits (@user_482391)

When plotting bio length by class, the red spike at zero tells the story. Real people write something, even if two words.

These patterns became the blueprint for engineered features.

Model Comparison Results



BiGRU-LSTM

Neural network that reads the bio word-by-word in both directions, combined with profile numbers

F1: 0.786 | Training: 14 min



CNN-BiLSTM

Neural network that scans the bio for short patterns, combined with profile numbers

F1: 0.789 | Training: 14 min



XGBoost (Winner)

Decision tree model on 37 profile numbers. No text reading.

F1: 0.808 | Training: 1 second

XGBoost: 87% accurate. 14 minutes faster to train than the neural networks.

- Why trees won:** "On tabular numeric data, **the signal lives in ratios and thresholds**, exactly what decision trees learn natively. Neural networks must reconstruct these splits." (*Shwartz-Ziv & Armon, Information Fusion 2022; Grinsztajn et al., NeurIPS 2022*)

Current Limitations

- Training data from 2018–2020 predates LLM-generated bots
- No social graph or full tweet history available
- English-dominant bios underrepresent other languages
- DOM scraping occasionally misses metadata

Future Roadmap

1

Fix DOM Scraping

Recover follower count and account age from hover cards, highest-value post-deployment change

2

Modern Dataset

Retrain on 2025–2026 data with LLM-era bots (current data is 2018–2020)

3

Chrome Web Store

Bundle model inside extension via ONNX, no backend server needed

4

Platform Expansion

Same architecture generalizes to Reddit, YouTube, Threads comments

Thank You!

Feel Free To Ask Any Questions